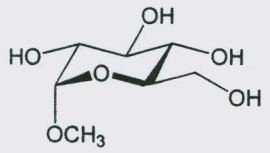
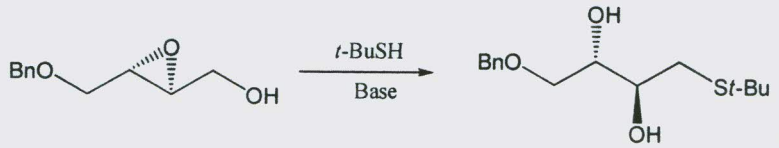
Organic Cumulative Exam
January 25, 2013

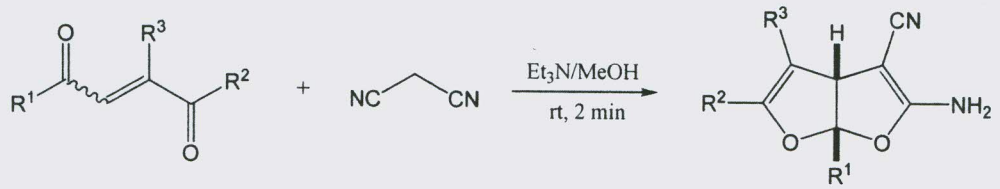
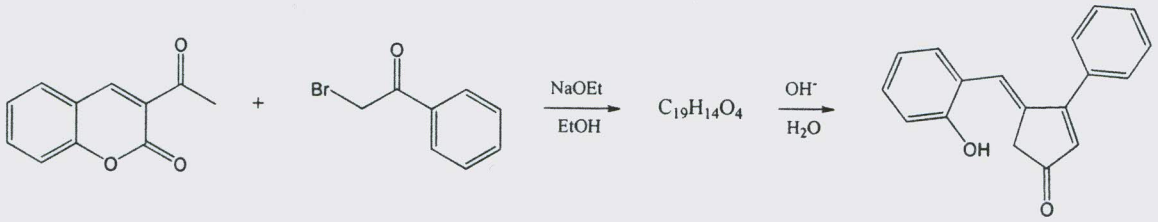
1. Determine the configuration of the following monosaccharide (15 points):
   (a) D- or L-? (b) α- or β-?



2. Write a mechanism for the following reaction. (15 points)



3. Write a mechanism for the following transformation (no need to explain stereochemistry). (30 points)



4. Provide the structure for the product after the first step and write a mechanism for the transformation over two steps. (40 points)